DPA 4099 — CLIP-ON INSTRUMENT MIC

Mounting, placement & EQ for the top 20 instruments and applications

Your kit: 4 × 4099 CORE+ **Extreme SPL** (yellow band)

Supercardioid · 2 mV/Pa · 152 dB peak · 80 Hz–15 kHz

Live Sound KB · companion to article mic-dpa-4099

The Mic & What “Extreme SPL” Means for You

The 4099 ships in two fixed-sensitivity variants, distinguished by a colored band. Yours are the **Extreme SPL** (yellow) — 10 dB less sensitive and a touch noisier than the Loud SPL, deliberately built to sit inside very loud sources without ever distorting.

The one rule that follows from this
On loud sources (brass, sax, drums, percussion, cranked amps) these are ideal and effectively bomb-proof. On delicate classical material played quietly — solo violin *pp*, harp, flute — you'll push the Q225/Wing pre harder and the 28 dB(A) floor starts to show. Still clean and usable; just gain-stage deliberately and get the capsule as close as placement allows.

Built-in character

A deliberate **+2 dB soft boost at 10–12 kHz** is baked in — that's the "air" up close. Don't stack another HF boost on top of it; if a source turns harsh, pull that region down instead. Supercardioid + close placement means constant **proximity effect** (always HPF) and high **placement sensitivity** (small moves swing the tone — find the spot before reaching for EQ).

Connection

MicroDot at the gooseneck end. The supplied XLR adapter makes it a normal 48 V phantom mic and carries a permanent **2nd-order 80 Hz low-cut** — perfect against handling/stage rumble. MicroDot also adapts to most pro wireless bodypacks. Always run it in the foam windscreen and rubber shock mount.

Spec	Loud SPL	Extreme SPL — YOURS
Band color	—	Yellow
Sensitivity	6 mV/Pa (-44 dB)	2 mV/Pa (-54 dB)
Max SPL (peak)	142 dB	152 dB
THD <1% up to	~133 dB	~137 dB
Self-noise (A-wt)	~23 dB(A)	28 dB(A) typ / 31 max
Dynamic range	~108 dB	109 dB
Pattern	Supercardioid	Supercardioid
Freq. range	80 Hz–15 kHz	80 Hz–15 kHz
HF lift	+2 dB @ 10–12 kHz	+2 dB @ 10–12 kHz

Gain note: expect ~10 dB more preamp gain than a typical condenser on the same source. Well within Q225 headroom — set to the loudest passage and ride.

Clip & Mount Ecosystem

The mic is universal — the **clip** makes it instrument-specific. One gooseneck + mic swaps between clips in seconds. All current MicroLock clips are backward-compatible with your CORE+ mics.

Clip	For	Mounts by
VC4099	Violin · viola · mandolin · banjo	Height-adjustable clamp on body edge (35–55 mm)
C-CLIP (4099C)	Cello	Clamps strings between bridge & tailpiece (curves out)
BC4099 (4099B)	Double bass	Clamps outer strings, bridge–tailpiece (curves in)
GC4099 (4099G)	Guitar · dobro · larger bodies	Sprung body clamp (35–122 mm depth)
STC4099 (4099S/T)	Sax · trumpet · trombone · bass clar.	Squeeze-to-fit bell/bow clamp
P-CLIP (4099P)	Piano (stereo pair)	Two holders seated in the frame
D-CLIP (4099D)	Drums · most percussion	Rim clamp, gooseneck flips up or down
U-CLIP (4099U)	Oboe · clarinet · flute · bassoon	Hook-and-loop strap, 90°-rotatable gooseneck
A-CLIP	Accordion	Dedicated accordion mount
CM-CLIP	Universal clamp	Music stands, rims, hardware
MS-CLIP	Mic-stand mount	3/8" + 5/8" thread
CS-CLIP	Cold shoe / camera	1/4" thread
G-MOUNT / XLR-MOUNT	Gooseneck or XLR-base stand use	Adds stand compatibility

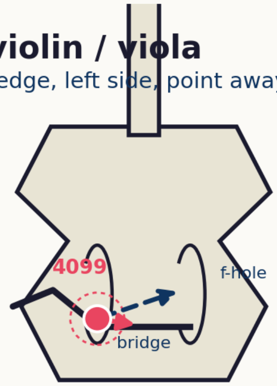
Open question — your clip kit
Mic count is confirmed (4 × Extreme SPL, yellow). Which clips you actually have on hand isn't logged yet. Tell me and I'll record it in the KB so the per-instrument picks below only ever point at gear you own.

Mounting — Strings

Bowed strings sit between bridge and tailpiece (cello, bass) or on the body edge (violin, viola). Aim is the whole game.

Strings — violin / viola

VC4099 on body edge, left side, point away from player's head



AIM AT BRIDGE
truest, balanced tone

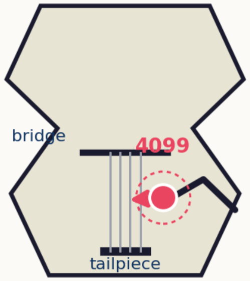
AIM AT F-HOLE
more output, duller/boxier

Viola = same VC4099 clip and logic, voiced darker.

Violin / viola. VC4099 on the body edge, **left side**, capsule pointed away from the player's head to dodge breath. **Bridge = truest tone; f-hole = louder but duller.** Viola same clip, darker voicing.

Cello & double bass

C-CLIP / BC4099 clamp the outer strings between bridge & tailpiece



Most NATURAL
below the bridge, between belly and strings

Most OUTPUT
angle toward an f-hole (accepts boxiness for gain)

Cello: holder curves OUT
Bass: holder curves IN

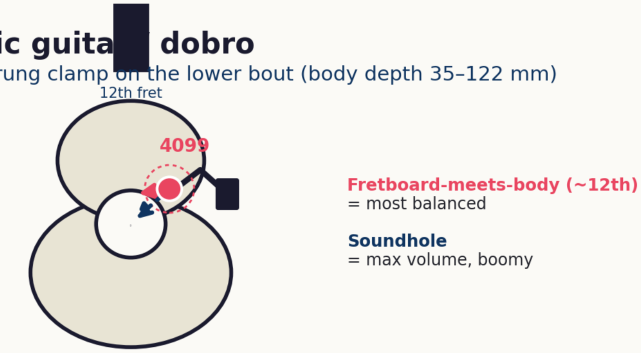
Cello / double bass. Outer strings, bridge–tailpiece. Cello holder curves **out**, bass curves **in**. Natural sound below the bridge; angle at an f-hole for max output. Extreme SPL loves hard pizz and slap.

Mounting — Fretted & Piano

Body-clamp for guitars; in-frame pair for piano.

Acoustic guita[redacted] dobro

GC4099 sprung clamp [redacted] the lower bout (body depth 35–122 mm)

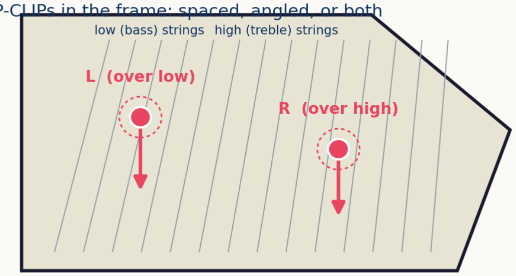


On stage, blend with the onboard pickup for gain-before-feedback. Mind piezo quack 1.5–2 kHz.

Acoustic guitar / dobro. GC4099 on the lower bout. **Fretboard-meets-body (~12th fret) = most balanced;** soundhole = max volume but boomy. On stage, blend with the onboard pickup for gain-before-feedback. Watch piezo quack 1.5–2 kHz.

Grand piano — 4099P stereo pair

Two P-CLIPs in the frame: spaced, angled, or both



Use the matched (white-dot, ±1 dB) pair + DAO4099 double cable. Classical: minimal EQ; check pair phase first.

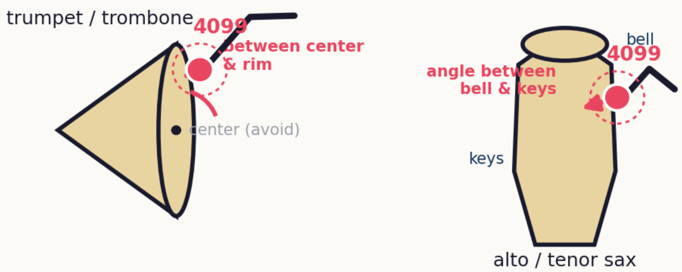
Grand piano. 4099P pair seated in the frame — spaced, angled, or both. One over the low strings, one over the high. Use the matched (white-dot) pair + DAO4099 double cable. This is your Simon & Garfunkel rig.

Mounting — Brass, Sax & Woodwinds

Bell-clip for brass and sax; strap-mount with distance for the woodwind family.

Brass & saxophone

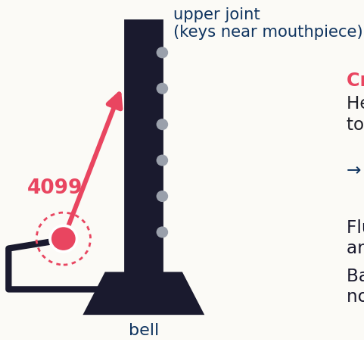
STC4099 on the bell — never straight into the center



Soprano sax: far from bell = warm, in front of bell = bite. Mutes are fine in place.

Woodwinds — clarinet / oboe / flute / bassoon

U-CLIP strap; beat the supercardioid's uneven timbre with distance



Create distance with the gooseneck
Head ABOVE the bell, twisted BACK toward the upper joint

→ covers the whole range evenly

Flute: near head joint but angled off the air jet.

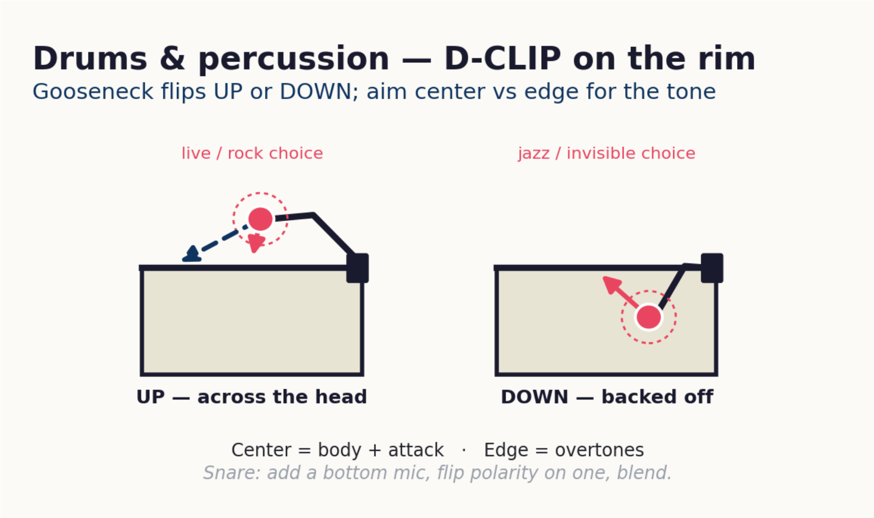
Bassoon: mic the tone holes, not the bell.

Brass & sax. STC4099 on the bell. **Never aim into the center** — position between center and rim (brass), or angle between bell and keys (alto/tenor sax). Soprano: far from bell = warm, in front = bite. Mutes are fine in place.

Woodwinds. U-CLIP. The supercardioid can give uneven timbre, so **create distance, place the head above the bell and twist back toward the upper joint.** Flute: near the head joint but angled off the air jet. Bassoon: mic the tone holes, not the bell.

Mounting — Drums & Percussion

D-CLIP on the rim. The gooseneck orientation sets the whole character.



Up = mic across the head, the live/rock choice — center for body & attack, edge for overtones. **Down** = backed off, lower-profile, the jazz and broadcast-invisible choice. Snare benefits from a top+bottom pair (flip polarity on one, blend). Congas, djembe, cajon, bongos all take the D-CLIP or CM-CLIP. Squarely the Extreme SPL wheelhouse.

EQ Starting Points — Top 20

The 4099 lives on acoustic and classical sources, so these default to **conservative** — not the aggressive non-classical defaults. **Whole-dB values only. Always HPF.** The built-in +2 dB at 10–12 kHz means you rarely need to add air — if anything, tame it. Console-neutral moves (map to Q225 4-band or Wing 6-band). Frequencies are starting points; placement does most of the work first.

#	Source	Clip	HPF	Move 1	Move 2	Move 3	Notes
1	Violin	VC4099	80 Hz	-3 @ 300 Q1.5 (box)	-2 @ 2.5k Q1.2 (scratch)	tame 10–12k if harsh	Aim at bridge
2	Viola	VC4099	70 Hz	-3 @ 250 Q1.5	-2 @ 2k Q1.2	—	Darker than violin
3	Cello	C-CLIP	50 Hz	-3 @ 200 Q1.5	-2 @ 400 Q1.2	+2 @ 4k if dull	Below bridge = natural
4	Double bass	BC4099	40 Hz	-4 @ 250 Q1.8	+2 @ 700 Q1.0 (pizz)	+2 @ 2.5k (slap)	Jazz: favor definition
5	Harp	U / CM	60 Hz	-2 @ 200 Q1.5	-2 @ 350 Q1.5	—	Quiet — mind noise floor
6	Acoustic guitar	GC4099	90 Hz	-4 @ 200 Q1.8 (boom)	-4 @ 1.8k Q1.8 (quack)	+3 @ 5k if dull	Quack only w/ pickup
7	Dobro / reso	GC4099	90 Hz	-4 @ 400 Q2.0	+3 @ 3k Q1.0 (bite)	LPF 12k if clangy	Bright by nature
8	Mandolin	VC4099	120 Hz	-4 @ 300 Q1.8	+3 @ 4k Q1.0	—	Percussive attack
9	Banjo	VC4099	120 Hz	-4 @ 400 Q1.8	-3 @ 1k Q1.5 (plink)	LPF 14k	Loud/bright — back out
10	Grand piano (pair)	P-CLIP	50 Hz	-3 @ 250 Q1.5	-2 @ 350 Q1.5	check pair phase	Classical: minimal
11	Flute	U-CLIP	200 Hz	-4 @ 400 Q1.5	+3 @ 5k Q0.8	notch breath	Angle off air jet
12	Clarinet	U-CLIP	120 Hz	-3 @ 350 Q1.5	-2 @ 1.5k Q1.2 (honk)	—	Above bell→upper joint
13	Oboe	U-CLIP	150 Hz	-3 @ 500 Q1.5	-2 @ 1.2k Q1.5	ease HF	Reedy — conservative
14	Bassoon	U-CLIP	60 Hz	-3 @ 300 Q1.5	+2 @ 2k Q1.0 (def)	—	Mic tone holes, not bell
15	Soprano sax	STC4099	100 Hz	-3 @ 400 Q1.5	-3 @ 2.5k Q1.5 (shrill)	—	Far from bell = warm
16	Alto / Tenor sax	STC4099	80 Hz	-4 @ 300 Q1.8	-3 @ 800 Q1.5 (honk)	+3 @ 5k Q1.0 (air)	Between bell & keys
17	Bari sax / bass clar	STC/U	50 Hz	-4 @ 250 Q1.8	-3 @ 700 Q1.5	+2 @ 4k	More bell distance
18	Trumpet	STC4099	80 Hz	-4 @ 300 Q2.0	-4 @ 1k Q1.8	+4 @ 5k Q1.0	Off-center of bell
19	Trombone / Fr. horn	STC4099	60 Hz	-4 @ 400 Q2.0	-4 @ 1.5k Q1.8	+3 @ 100 shelf	Give slide/bell room
20	Hand perc / drum spot	D-CLIP	per src	center=body, edge=air	top+btm pair, flip pol	gate carefully	Extreme SPL wheelhouse

Move format: ***gain @ freq Q*** *(e.g. "-4 @ 300 Q1.8" = cut 4 dB at 300 Hz, Q 1.8). For loud sources (16–20) your aggressive defaults from eq-starting-points apply more than the conservative string/woodwind approach. For 1–14, stay subtractive — the 4099's job is to disappear.*

Field Cheat-Sheet

Always	HPF every channel (proximity). Foam windscreen + shock mount on. 80 Hz low-cut in the XLR adapter handles rumble.
Don't	Add HF air — the mic already has +2 dB at 10–12 kHz baked in. If harsh, cut there instead.
Placement first	Narrow supercardioid pattern swings tone hard with small moves. Nail the spot before EQ.
Aim rules	Strings: bridge = true, f-hole = loud/dull. Brass/sax: off-center of bell, never dead center. Guitar: 12th fret = balanced.
Gain (Extreme SPL)	~10 dB hotter pre than a normal condenser. Loud sources: set it and forget it. Quiet classical: get close, watch the 28 dB(A) floor.
Two-mic tricks	Snare and kick both reward a second 4099 on the opposite side — flip polarity on one, blend to taste.

Sources

DPA Microphones — 4099 CORE+ product page & manual; "How to mount the 4099 on various instruments" (Bo Brinck); CORE+ campaign specs; Extreme SPL datasheet (2 mV/Pa, 152 dB, 28 dB(A), 109 dB DR); Sound On Sound 4060/4099 CORE review. EQ approach per Brian Lloyd's documented philosophy (KB: eq-starting-points). Photos/videos for each instrument: dpamicrophones.com mic-university.

Open questions logged in the KB

(1) Which clips are in your kit for the four mics? (2) Any matched/white-dot pairing for stereo piano/strings? (3) Wireless adapters on hand, or all wired via XLR into the Q225/Wing? (4) Bari-sax / harp clip from kit or borrowed?